BENCHMARK REPORT

Listening across languages. Seeing with precision.

Songa model evaluation | Speech recognition and visual segmentation

Songa is a travel companion for East Africa, designed around the way people speak and the places they explore. We evaluated the speech recognition behind Speak and Journal, and the segmentation behind Look, to understand how reliably these components support the experience.

Our speech benchmark includes 23 team-collected Kenyan recordings with human-verified references, including 20 Swahili-English code-switched clips. A further 27 AfriSpeech recordings assess accented English. For vision, we compared self-hosted SAM 3 with hosted SAM 3.1 on the same 200 COCO images.

16.2%
Word error rate
Songa recordings · 95% CI 13.5-19.4%

88.0%
Swahili words matched exactly
Within the 20 code-switched recordings

94.5%
Object presence recall
Self-hosted SAM 3 · hosted SAM 3.1: 88.6%

0.76 s
Median tap response
Self-hosted SAM 3 · warm requests

What the results show

Speech: Swahili was largely retained in the tested code-switched recordings. Word-boundary changes, substitutions and language-switch errors remain relevant to meaning preservation.

Vision: Self-hosted SAM 3 had higher detection recall and box/mask AP, with lower warm latency. Hosted SAM 3.1 had lower count error and a higher share of taps meeting the mask-overlap thresholds.

> **Scope of the evidence**
> This is a component benchmark. It does not measure the accuracy of final translations, visual explanations, diary generation or navigation.

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01 / STUDY DESIGN

What we measured

A single evaluation run on Modal, 3 November 2026. Total wall time: 23.2 minutes.

Speech set	Clips / words	Recording conditions
Songa recordings	23 / 616	374.1 s total; 3.5-23.0 s per clip; mean 16.3 s.
Health complaints	19 / 597	Scripted, naturalistic Swahili-English; smartphone audio in quiet indoor settings.
Short triage phrases	4 / 19	Laptop microphone; 2 English, 1 Kiswahili and 1 code-switched phrase.
AfriSpeech-200 sample	27 / 445	242.9 s total; mean 9.0 s. 14 clinical and 13 general-domain clips.

The health complaints span 15 clinical categories. AfriSpeech contributes three clips from each of nine language groups, represented by 11 language-country accent labels. All 50 clips had human-verified reference transcripts and completed successfully.

Speech request and scoring protocol

Input: Each clip was converted to mono, 16-bit PCM WAV at 24 kHz, then transcribed using Muse Voice Transcribe (**muse-voice-transcribe-1.0**). English clips received an English language hint; Kiswahili and code-switched clips received no hint.

Execution: Three clips ran concurrently, with up to four retries on a failed request. Request latency includes conversion and the network round trip from Modal to the model API, but excludes queueing before a concurrency slot becomes available.

Text normalisation and aggregation

Text was Unicode-normalised and lowercased; digits were written as words; hyphens became spaces; punctuation and apostrophes were removed; whitespace was collapsed. For AfriSpeech, spoken punctuation was also removed from transcripts when the reference did not contain the same word.

Measure	How to read it
Word error rate (WER)	Word substitutions + deletions + insertions, divided by reference words. Lower is better.
Character error rate (CER)	The corresponding character-level error rate, with spaces included. Lower is better.
Pooled results	Total errors divided by total reference units. Longer clips contribute more weight.
95% confidence interval	1,000 bootstrap resamples of whole clips, seed 7; 2.5th and 97.5th percentiles. Reported for groups of at least five clips.

Clip-level resampling preserves within-recording dependence. Speaker IDs were unavailable, so the intervals do not account explicitly for repeated speakers across clips.

02 / SPEECH ACCURACY

Transcription across both datasets

Muse Voice Transcribe | Normalised, pooled error rates

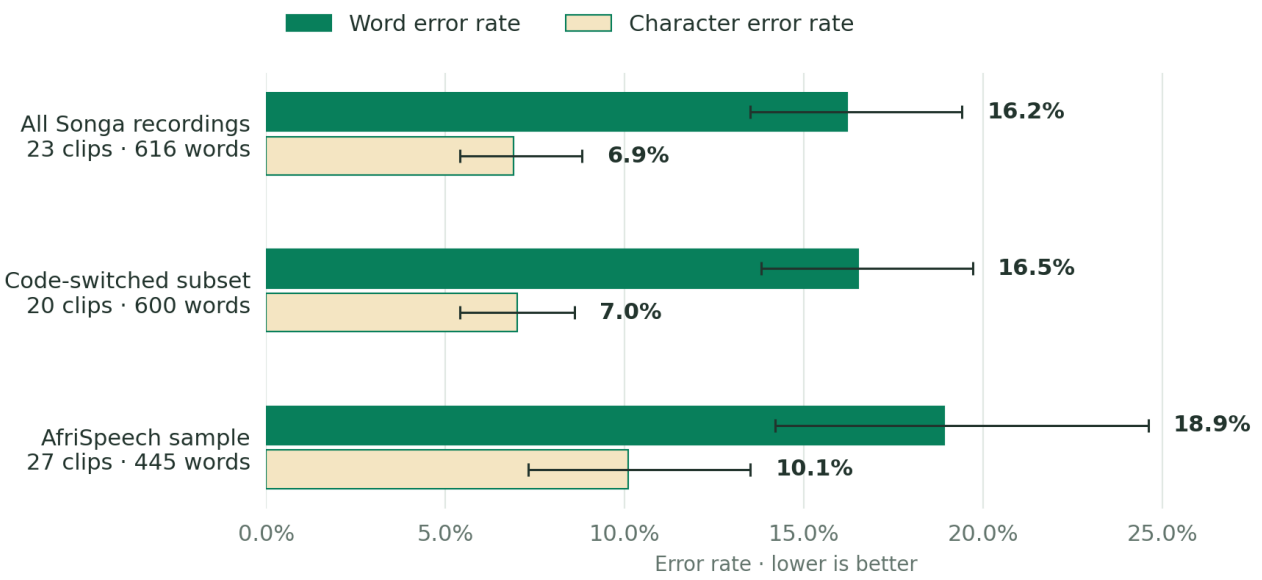


Figure 1. WER and CER with 95% bootstrap confidence intervals. The code-switched subset is part of the full Songa set, not an independent dataset.

Evaluation group	Clips / words	WER	CER
All Songa recordings	23 / 616	16.2% 13.5-19.4	6.9% 5.4-8.8
Code-switched subset	20 / 600	16.5% 13.8-19.7	7.0% 5.4-8.6
English phrases	2 / 11	9.1%	7.1%
Kiswahili phrase	1 / 5	0.0%	0.0%
AfriSpeech accented English	27 / 445	18.9% 14.2-24.6	10.1% 7.3-13.5

Ranges beneath the pooled estimates are 95% confidence intervals. The English and Kiswahili phrase groups are too small to support a general performance conclusion.

Reading the two error measures

CER is lower than WER in both datasets. Small spelling changes and merged or split words contribute to this gap, but the gap alone does not show whether the intended meaning was preserved.

Normalisation effect: Mean per-clip WER fell from 26.3% to 17.2% for Songa and from 44.3% to 19.8% for AfriSpeech. These are unweighted clip means and differ from the pooled headline values above.

03 / CODE-SWITCHING

How well did each language survive?

20 code-switched clips | 600 reference words: 500 tagged Swahili and 100 tagged English

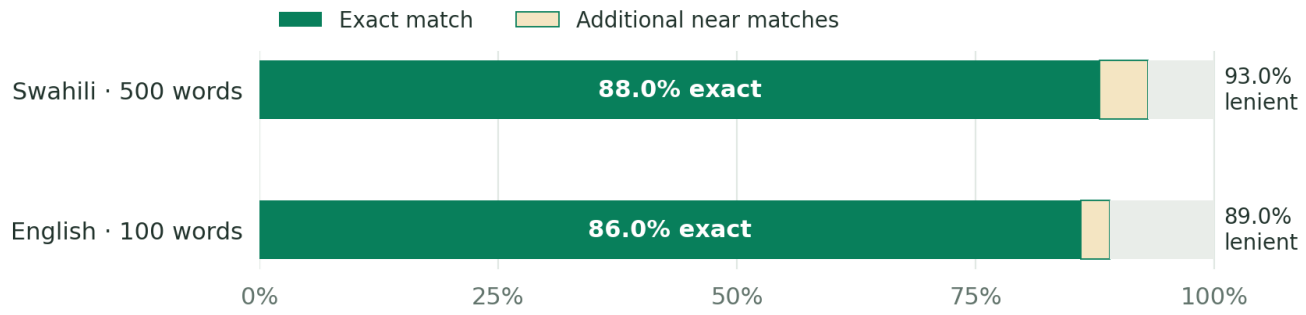


Figure 2. Exact word recovery and the additional matches admitted by the lenient spelling rule. Grey represents words outside that rule.

Reference group	Exact recall	Lenient recall
Swahili, code-switched subset	88.0% (84.8-90.9)	93.0% (90.1-95.6)
English, code-switched subset	86.0% (75.5-94.1)	89.0%
Swahili, all Songa recordings	88.1% (85.0-91.0)	93.1% (90.3-95.5)
English, all Songa recordings	87.4% (76.9-94.4)	90.1%

Parentheses show available 95% confidence intervals. Lenient recall allows a substitution within one character, or two for words of eight or more letters; it is not a measure of semantic correctness.

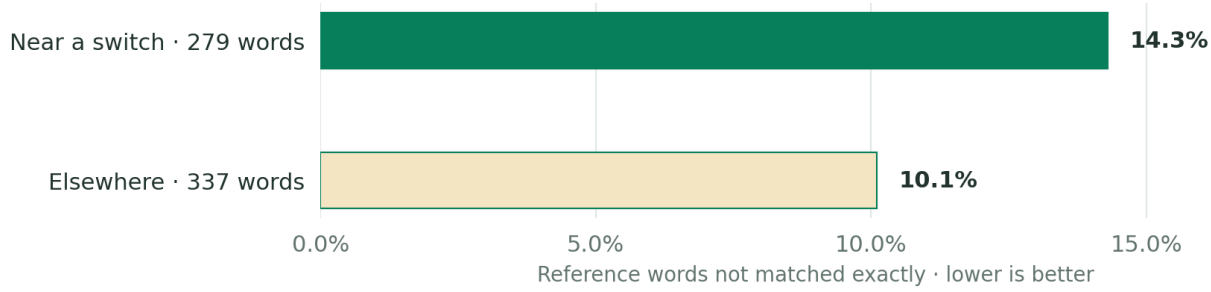


Figure 3. Reference-word mismatch rate near a language boundary versus elsewhere across Songa recordings. The comparison is descriptive; no confidence interval was estimated.

Switch effect: Mismatch rates were 14.3% near a switch and 10.1% elsewhere: a 4.2 percentage-point gap, or about 1.4 times the rate. The window covers the two words on each side of a language boundary.

Swahili error breakdown: Across all 505 Swahili-tagged words, 445 matched exactly. The 60 misses comprised 25 near-miss spellings, 30 other substitutions and 5 deletions. Three words (0.6%) were substituted with a word tagged as English.

Language tags use English word frequency with a 68-word Swahili override list. Borrowed words may fall on either side; this affects recall and switch-window metrics, but not overall WER or CER.

04 / SPEECH BEHAVIOUR

Where the errors occur

Illustrative transcription errors and measured request latency

Pattern	Reference → transcript	What it shows
Word boundaries	ni tangu juzi → nitangujuzi	A merge can incur several word errors with few character changes.
Prefix changes	natokwa → tokwa naona → ona	Short character changes can alter word form.
A mixed-language word	Naweza kubook appointment? → Naweza kubuka pointment.	2 of 3 words wrong: 66.7% WER, 16.0% CER.
Translation within transcription	Amekuwa na fever → ... na homa	One observed English-to-Swahili substitution.
Domain vocabulary	hyaline degeneration → high line degeneration	A technical term was replaced by common words.
Dates and numbers	21/22/20 → twenty fourth twenty second twenty	Formatting and number-word interpretation affect scoring.

The sample also contained a non-Latin character inside one Swahili word. Such output merits review; automatically deleting characters could introduce a further transcription error.

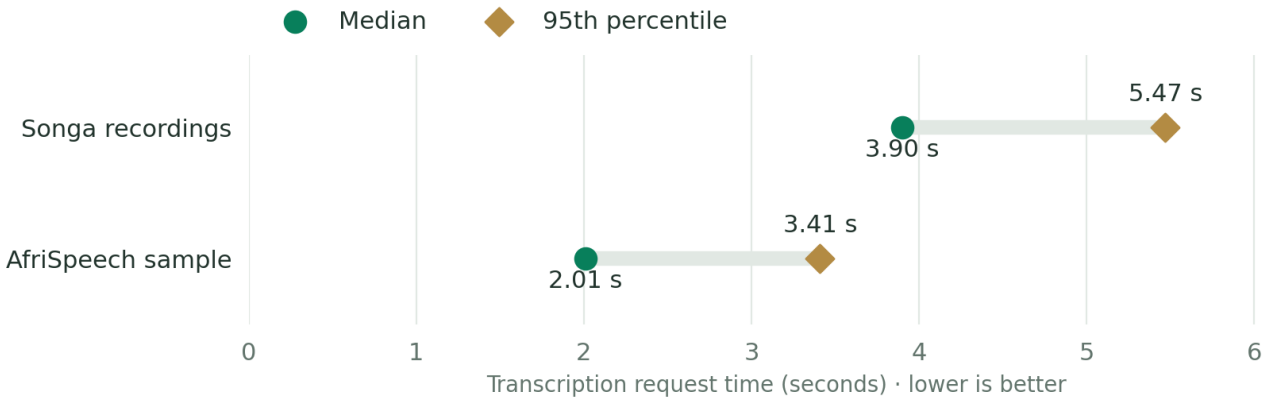


Figure 4. Transcription latency from Modal. These timings exclude translation, speech playback and the phone-to-gateway network leg.

Songa clips averaged 16.3 seconds of audio, compared with 9.0 seconds for AfriSpeech. The four short triage phrases took 1.2-1.6 seconds to transcribe. Six of 23 Songa clips had WER at or below 10%; health-complaint WER ranged from 6.7% to 29.4%.

Implication for Speak and Journal
Showing the transcript early gives users something to review while later stages run. This benchmark measures the transcript itself; it does not establish the fidelity of the final interpretation or diary entry.

05 / ACCENTED ENGLISH

Performance varies across the sample

AfriSpeech-200 | 27 clips across 11 language-country accent labels

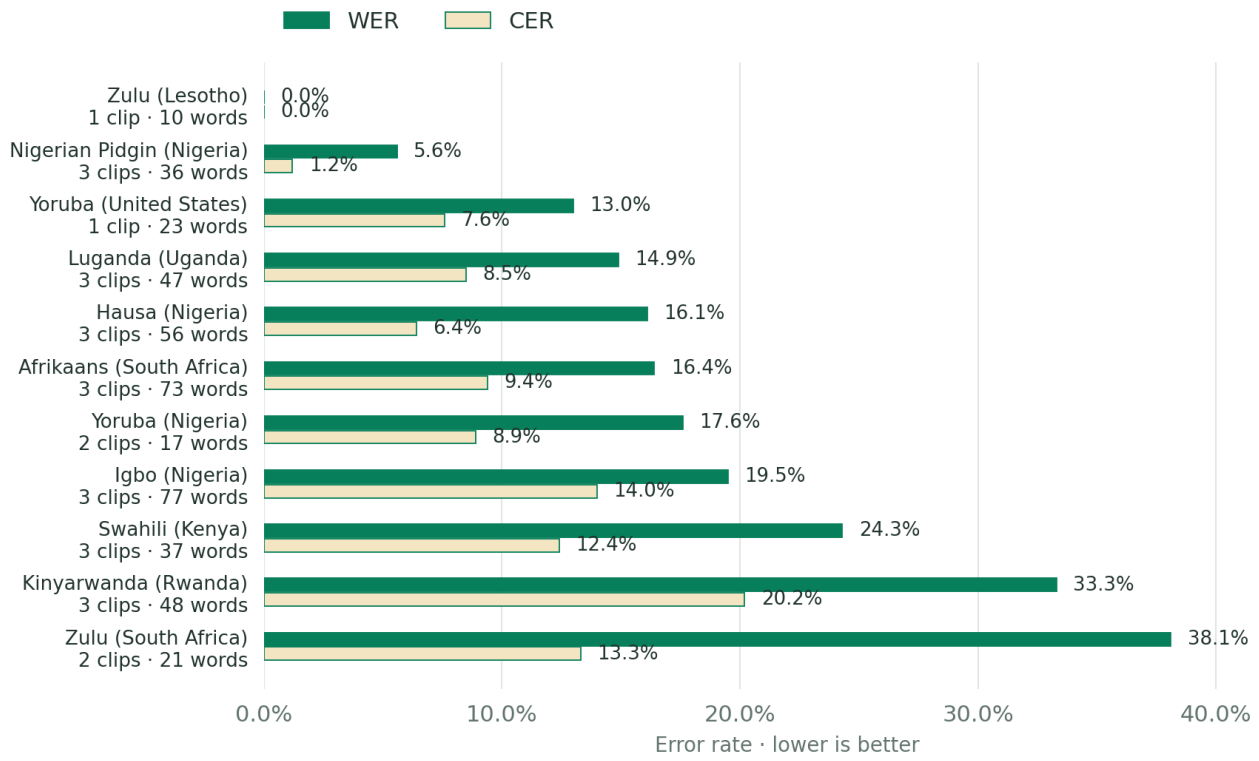


Figure 5. Pooled WER and CER by accent label. Each row contains only 1-3 clips and 10-77 words; the ordering should not be interpreted as a reliable accent ranking.

Domain	Clips	Words	Pooled WER
Clinical text	14	206	18.0%
General text	13	239	19.7%
All AfriSpeech	27	445	18.9%

Across AfriSpeech, English word recall was 86.5% exact (95% CI 81.4-90.3%) and 89.7% under the lenient spelling rule. Errors included technical vocabulary, names, dates and number expressions.

What this sample supports

The result adds evidence on accented English beyond the team recordings. The 27 clips were selected as three per language group and are not established as a random sample of the full corpus.

06 / VISION STUDY DESIGN

A shared vision benchmark

Self-hosted SAM 3 on Modal and hosted SAM 3.1 through the Meta Model API

200

COCO val2017 images
Sampled with seed 7

358

Interior object taps
Same points tested on both backends

The 1,003 text prompts comprised 603 present-category prompts and 400 absent-category prompts. Each photo contributed every annotated category present plus two randomly selected absent categories. Both backends received identical JPEG bytes at the original resolution.

Configuration	Self-hosted SAM 3	Hosted SAM 3.1
Model	SAM 3 open weights	SAM 3.1 API
Execution	Modal GPU: L4 requested, A10 permitted; actual GPU type not logged	Hosted inference; requests sent from Modal
Text prompts	One batched call per photo; internal score floor 0.05	One request per prompt; four concurrent requests
Point prompts	Four taps in flight sharing one GPU	Four taps in flight
Duplicate removal	Box IoU 0.60, per prompt	Box IoU 0.60, per prompt
Scoring	All retained scores used for AP	Missing confidence scores assigned 1.0 for AP

How the tests were constructed

Presence and counting: The operating threshold was 0.35. A category counted as present if its highest-scoring detection reached that threshold. Count error compared detections above the threshold with annotated non-crowd instances.

Tap selection: Up to two eligible non-crowd objects per image were selected at random. Objects had to occupy at least 1,024 pixels². The tap was placed at the point deepest inside the ground-truth mask, giving a clear interior target.

Timing: The GPU was warmed before measurement. Photos ran sequentially; per-photo time covered the first prompt request through the final answer. Tap time began when a concurrency slot became available.

This compares two complete deployment paths. Model version, request batching, network overhead and returned detections differ, so the measurements do not isolate a model-only effect.

07 / DETECTION QUALITY

Self-hosted SAM led detection

Presence at confidence ≥0.35 | COCO AP averaged over IoU thresholds 0.50-0.95

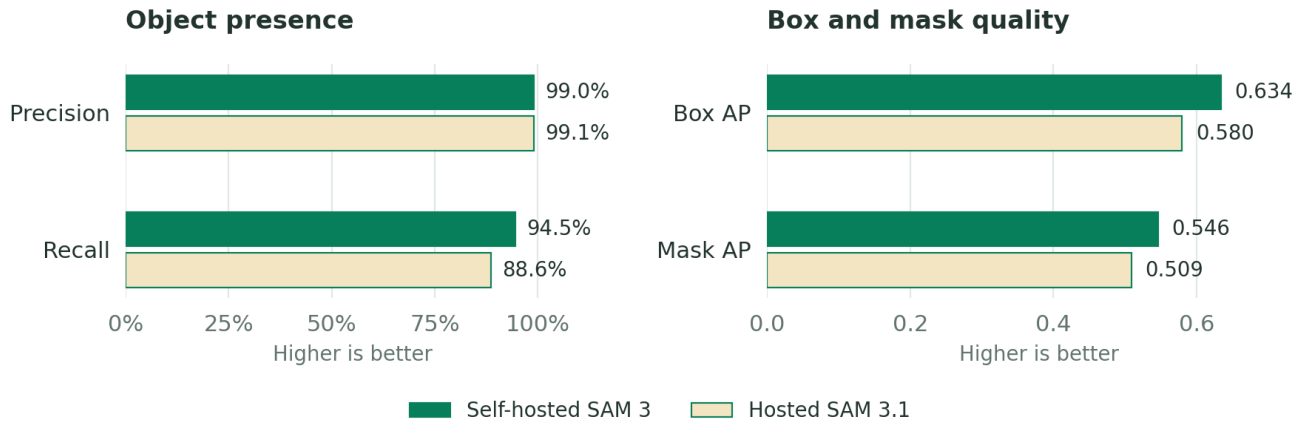


Figure 6. Presence and average-precision measures for the two backends. Presence rates and AP use separate axes; higher is better in each panel.

Measure	Self-hosted SAM 3	Hosted SAM 3.1
Present prompts found / missed	570 / 33	534 / 69
Absent prompts flagged / rejected	6 / 394	5 / 395
False-positive rate on absent prompts	1.50%	1.25%
Presence precision / recall	99.0% / 94.5%	99.1% / 88.6%
Presence accuracy	96.1%	92.6%
Box AP / AP50 / AP75	0.634 / 0.822 / 0.682	0.580 / 0.739 / 0.631
Box AR100	0.737	0.618
Mask AP / AP50 / AP75	0.546 / 0.809 / 0.597	0.509 / 0.733 / 0.569
Mask AR100	0.638	0.550
Failed prompts / failed taps	0 / 0	0 / 0

AP summarises precision and recall across detection scores. AP50 and AP75 use fixed overlap thresholds; AR100 is average recall with up to 100 detections per image. Absent-category detections contribute false positives.

Detection trade-off

The self-hosted path missed 33 of 603 present-category prompts, compared with 69 for the hosted path. Both returned few false detections on absent-category prompts. These are annotation-based results; the flagged images were not manually reviewed.

08 / INTERACTION QUALITY

Tap isolation and counting

Tap masks are compared with the selected ground-truth object; empty outputs score zero.

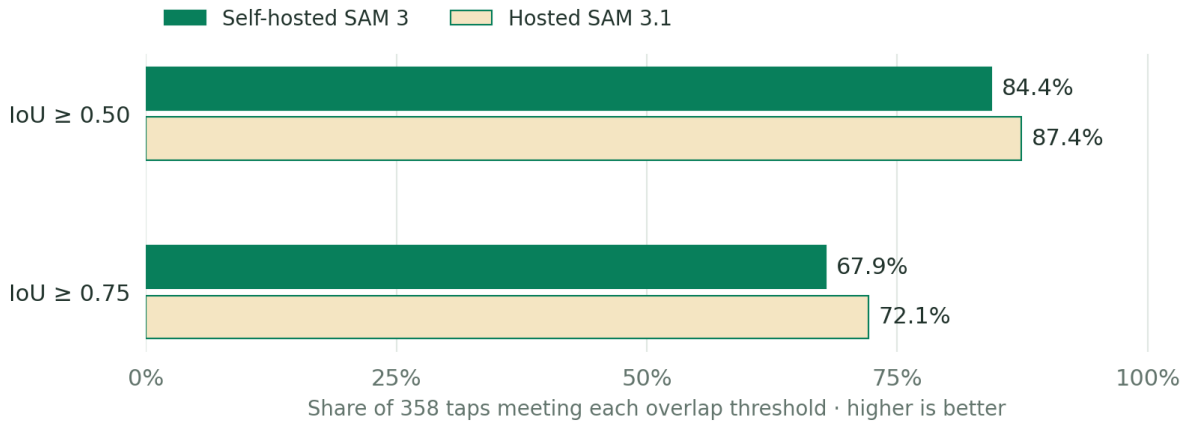


Figure 7. Hosted SAM 3.1 met both tap-overlap thresholds more often. A mask IoU of 1.0 is a perfect overlap; 0 is no overlap.

Tap measure	Self-hosted SAM 3	Hosted SAM 3.1
Mean / median IoU	0.735 / 0.846	0.762 / 0.855
Very low overlap: IoU <0.10	18 taps	12 taps
Better by >0.10 IoU	8 taps	20 taps

On 330 of 358 taps, the two backends differed by no more than 0.10 IoU. Low-overlap results were most often associated with people, chairs and beds, where the selected region may be a part rather than the full object.

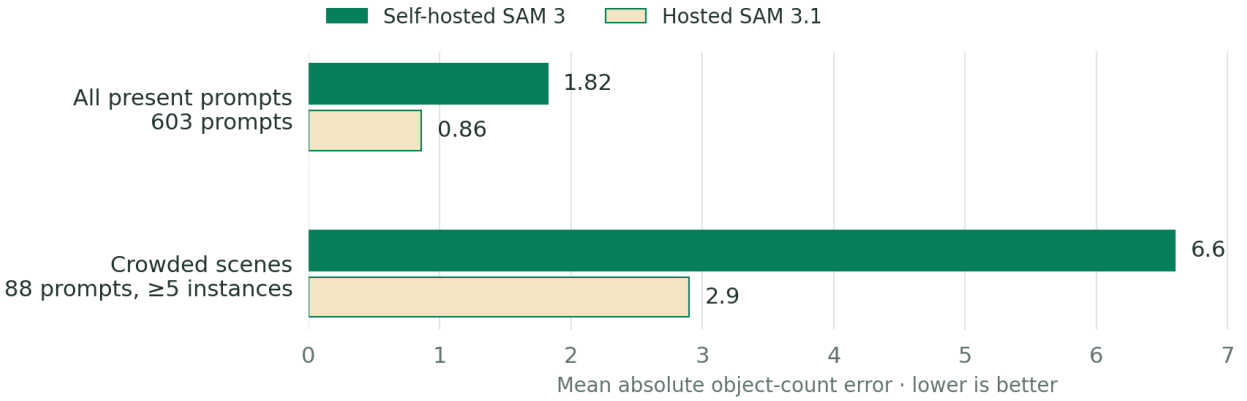


Figure 8. Count error is the mean absolute difference from annotated non-crowd instance counts. The crowded-scene subset contains prompts with at least five instances.

Direction of error: Mean absolute count error was 1.82 for self-hosted SAM and 0.86 for hosted SAM. Of 603 present prompts, the self-hosted path over-counted on 269 and under-counted on 47; the hosted path over-counted on 108 and under-counted on 128.

09 / RESPONSIVENESS

A clear warm-latency advantage

All segmentation timings were measured from Modal, with GPU cold start excluded.

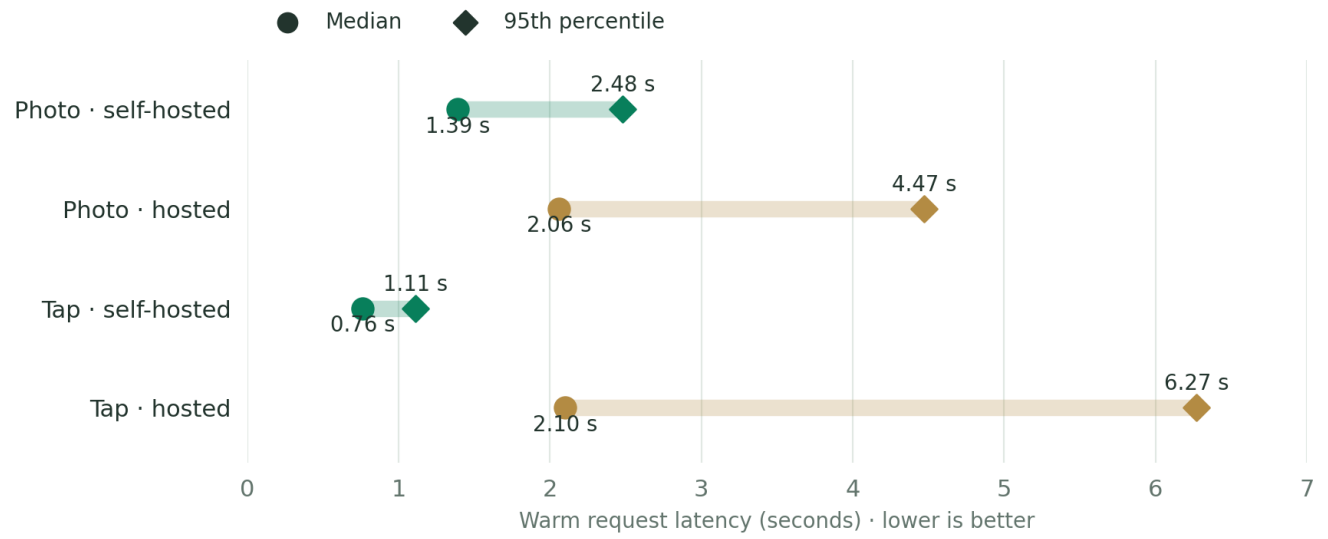


Figure 9. Median and 95th-percentile request latency. Green denotes self-hosted SAM 3; gold denotes hosted SAM 3.1. Circles show medians and diamonds show the 95th percentile.

Request	Backend	Median	p95	Mean
Photo	Self-hosted	1.39 s	2.48 s	1.54 s
Photo	Hosted	2.06 s	4.47 s	2.38 s
Tap	Self-hosted	0.76 s	1.11 s	0.83 s
Tap	Hosted	2.10 s	6.27 s	2.57 s

Tail latency is the largest difference
At the 95th percentile, hosted tap requests took 6.27 seconds versus 1.11 seconds for self-hosted SAM. This is a measured difference between the tested request paths, not an estimate of full phone-to-answer latency.

Execution context

The evaluation GPU took 2 minutes 35 seconds to boot and load weights before timing began. Photo requests used different batching patterns across backends, and the four concurrent GPU taps shared one container. Network and scheduling conditions reflect one run.

Recorded run usage: 23.2 minutes of wall time and $0.94 in Modal usage for CPU orchestration and GPU processing. Meta Model API usage was not measured, so this is not a complete inference-cost estimate.

10 / FINDINGS & NEXT STEPS

What the benchmark supports

Measured component performance, with a focused path to broader validation

Finding	Product implication
Speech preserved most Swahili words in this sample.	Continue testing code-switched input without a Kiswahili hint. Make transcripts reviewable before their content feeds interpretation or diary generation.
Self-hosted SAM led detection recall, AP and warm speed.	It is a strong candidate for responsive object discovery when GPU capacity is available. Validate the same advantage on Look's actual scene types.
Hosted SAM had lower count error and stronger tap overlap.	Keep counting and tap isolation as separate acceptance criteria when choosing a deployment path.

Limits on interpretation

Speech coverage. The team recordings are scripted health complaints and short phrases recorded in quiet settings. Speaker identities were not logged, and street noise, spontaneous travel dialogue, Sheng and coastal Swahili were not covered. Language tags are approximate, and performance was measured once.

Vision coverage. COCO category names and centrally placed taps are a limited proxy for menus, signs and markets. The comparison changes model version and request pattern; GPU type was not logged. Segmentation metrics have no confidence intervals.

System coverage. The run does not evaluate intent preservation after translation, Muse Wonder's explanations, diary quality, allergen guidance, navigation, cold-start fallback or phone-level response time.

Priority evaluations

1. Expand speech to at least 100 clips from 10 or more identified speakers, including spontaneous and noisy travel interactions.
2. Score end-to-end meaning preservation, especially negation, numbers, names and user needs; repeat model calls to quantify variability.
3. Test menus, signs and market scenes with concept prompts, natural taps and independent annotations.
4. Tune duplicate suppression and count thresholds, then measure cold starts, fallback behaviour, device latency and complete inference cost.

Dataset and model attribution

AfriSpeech-200 clips and transcripts: © Intron Health, CC BY-NC-SA 4.0, used for non-commercial evaluation. COCO annotations: COCO Consortium, CC BY 4.0; image rights remain with their owners. SAM 3 weights are distributed under Meta's SAM License. Hosted SAM 3.1 and Muse Voice Transcribe are used through the Meta Model API under its terms.

APPENDIX / SONGA RECORDINGS

Per-clip speech results

Sorted by WER | Descriptive recording labels | Latency is transcription request time

Recording	Language	WER	CER	Latency
English triage phrase 13	EN	0.0%	0.0%	1.31 s
Kiswahili triage phrase 23	SW	0.0%	0.0%	1.56 s
Fever and headache	SW-EN	6.7%	2.2%	3.93 s
Emotional distress	SW-EN	7.7%	2.0%	5.09 s
Eye exposure	SW-EN	9.7%	2.5%	4.00 s
Diarrhoea and vomiting	SW-EN	10.0%	7.5%	4.38 s
Child described	SW-EN	11.1%	9.3%	6.09 s
Sudden change in a relative	SW-EN	11.1%	3.3%	4.72 s
Pain when urinating	SW-EN	13.3%	8.8%	3.87 s
Injury and ongoing bleeding	SW-EN	13.8%	6.1%	3.90 s
Chest discomfort	SW-EN	16.1%	8.5%	5.26 s
English triage phrase 19	EN	16.7%	12.9%	1.44 s
Diabetes and missed food	SW-EN	16.7%	4.1%	3.94 s
Persistent cough	SW-EN	17.2%	6.7%	3.90 s
Head injury uncertainty	SW-EN	17.9%	6.7%	3.76 s
Pregnancy bleeding	SW-EN	18.8%	7.2%	3.52 s
Medication refill	SW-EN	19.4%	9.6%	3.29 s
Mild respiratory symptoms	SW-EN	20.7%	3.9%	3.33 s
Symptoms after medication	SW-EN	22.6%	12.1%	3.91 s
Dental pain and swelling	SW-EN	24.1%	6.8%	3.41 s
Difficulty breathing	SW-EN	25.0%	16.6%	4.10 s
Abdominal pain	SW-EN	29.4%	9.0%	5.50 s
Appointment triage phrase 30	SW-EN	66.7%	16.0%	1.20 s

Language: EN = English; SW = Kiswahili; SW-EN = code-switched Swahili-English. All recordings other than the four labelled triage phrases are scripted health complaints.

The 66.7% WER result comes from a three-word phrase, so two word errors dominate its score. Dataset-level results pool reference words rather than averaging this table’s percentages.